Reload the local package database, as follows:

```
sudo apt-get update
```

Install the latest version of MongoDB by using the following command:

```
sudo apt-get install mongodb-org
```

Issue the following command to start MongoDB:

```
sudo service mongod start
```

Once you have installed and started MongoDB, you can start the Mongo executable by issuing the following command:

```
$ mongo
MongoDB shell version: xxx
connecting to: test
```

By default, the MongoDB shell will be connected to the test database.

```
> db
test
```

The commands given in Figure 1 can be used for four basic CRUD (create, read, update, delete) operations.

In Figure 1, 'post' is a document that is getting saved to the 'article' collection in the database. The *find* command can be used to query a collection. MongoDB will not permanently create a database until you insert data into it, and it will create a collection implicitly upon its first use. So it is not required to create a collection before inserting data. Every document stored in MongoDB will have an *_id* (default type is ObjectID) key, which will automatically be inserted in the document, whose value will be unique for a particular collection.

For updates, the *db.collection.update()* and the *db.collection.save()* methods are available. These methods update a single document, by default. With the *multi* option, *update()* can update all documents in a collection that match a query.

```
· db.article.update({title:"Introduction to
MongoDB"},{$set:{date:"2 Feburary 2016"} })
```

To remove documents from a collection, use the *db.collection.remove()* method, and you can also pass a query criteria to determine which documents to remove.

MongoDB supports horizontal scalability and high throughput by sharding - a method for storing data across multiple machines. Sharding is done to increase available RAM and disk space, reduce load on a server, read or write data with better throughput, etc. In sharding, data is distributed across multiple servers and the ordered collections are saved across

Figure 1: Snapshot of CRUD operations in MongoDB

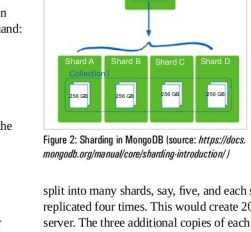

Figure 2: Sharding in MongoDB (source: *https://docs.mongodb.org/manual/core/sharding-introduction/)*

many machines. Shards are further replicated to allow failover. Each shard processes fewer operations and stores less data as the server grows; thus the capacity and throughput of the cluster will be increased horizontally. For example, a large collection could be split into many shards, say, five, and each shard in turn may be replicated four times. This would create 20 units of a MongoDB server. The three additional copies of each shard server will act as failover units.

In Figure 2, if a database has a 1 terabyte data set, and there are four shards, then each shard might hold only 256GB of data. If there are 40 shards, then each shard might hold only 25GB of data and thus the load on each shard will decrease. The MongoDB sharded cluster has shards, query routers and config servers. Shards store the data, and provide high availability and data consistency. Query routers (mongos instances) direct client operations to the appropriate shards. Config servers store the cluster's metadata, which is a mapping of the cluster's data set to the shards.

While MongoDB is a database which supports horizontal scalability, by adding more and more nodes to a cluster, it can give faster results for big data sets, but it should be avoided if the system requires lots of transactions. It is a more read-intensive system. The security options are also limited to the database level. The option to grant permissions at the collection level is not available. END

References

[1] *https://docs.mongodb.org/manual/mongo/*
[2] MongoDB – The Definitive Guide by O'Reilly
[3] *https://docs.mongodb.org/manual/reference/sql-comparison/)*

By: Surabhi Dwivedi

The author is working in a premier R&D organisation. Her interests are data mining, Big Data, database technologies, data modelling, Linux systems administration, open source technologies, etc. You can contact her on *dwivedisurabhi7@gmail.com*